Benchmarking Deep Reinforcement Learning Algorithms with Potential-Based Reward Shaping on LunarLander

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***Abstract*—This study benchmarks DQN, PPO, and A2C algorithms on the discrete LunarLander environment under four reward conditions: no shaping, distance potential, angle potential, and a combined potential. The objective is to identify which algorithm and reward-shaping combinations achieve the fastest learning without compromising final performance, while ensuring that policy optimality is maintained under Potential-Based Reward Shaping (PBRS).**

I. Project Description

Reinforcement learning (RL) agents generally require extensive interactions with the environment to acquire effective behaviors. Reward shaping provides additional guidance by modifying the reward signal; however, if not carefully designed, it can interfere with the optimal policy. Potential-Based Reward Shaping (PBRS) addresses this issue by incorporating a shaping bonus:

$$F(s, s') = \gamma\Phi(s') - \Phi(s), \tag{1}$$

which is guaranteed to preserve optimal policies [1]. Inspired by prior work [2], this project evaluates DQN [3], PPO [4], and A2C [5] from Stable-Baselines3 under the original reward and three hand-designed PBRS configurations, identifying which algorithm–shaping combinations improve learning efficiency without distorting optimal behavior. We hypothesize that PBRS will enhance sample efficiency more for DQN than for PPO and A2C. This is because DQN, being off-policy, updates value estimates from shaped rewards stored in its replay buffer, while PPO and A2C, as on-policy methods, only use rewards from rollouts that are discarded after each update. We will test this by comparing timesteps required to reach a fixed reward threshold and the area under the learning curve, reporting mean and standard deviation for each algorithm-reward configuration across different random seeds.

II. State Space

The observation space is a continuous 8-dimensional vector:

$$o = [x,\ y,\ \dot{x},\ \dot{y},\ \theta,\ \dot{\theta},\ \ell_L,\ \ell_R]^T \in \mathbb{R}^8, \tag{2}$$

where (x, y) is the lander's centre-of-mass position, $(\dot{x}, \dot{y})$ are linear velocity components, $\theta \in [-\pi, \pi]$ is the orientation angle, $\dot{\theta}$ is angular velocity, and $\ell_L, \ell_R \in \{0, 1\}$ are binary leg ground-contact indicators.

III. Action Space

The discrete action space is $A = \{0, 1, 2, 3\}$ ($|A| = 4$):

- $a = 0$: no operation.
- $a = 1$: fire left orientation engine.
- $a = 2$: fire main engine.
- $a = 3$: fire right orientation engine.

Using the discrete variant allows DQN, PPO, and A2C to be evaluated under identical environment conditions. To improve reproducibility, experiments use the no-wind setting of LunarLander.

IV. Proposed Solution

Three algorithms (DQN, PPO, A2C) are each tested across four reward shaping configurations, yielding $3 \times 4 = 12$ experimental conditions. For all PBRS conditions, the shaped reward is:

$$r'(s, a, s') = r(s, a, s') + \lambda\big[\gamma\Phi(s') - \Phi(s)\big], \tag{3}$$

with shaping coefficient $\lambda = 1.0$ and $\Phi(s_{\text{terminal}}) = 0$. The shaping coefficient and combination weights are fixed across algorithms to isolate the effect of potential design under a controlled setting. Potential values will be normalized using fixed scale factors: $\tilde{\Phi}_{\text{dist}}(s) = -\min(1, \sqrt{x^2 + y^2}/d_{\max})$ and $\tilde{\Phi}_{\text{angle}}(s) = -|\theta|/\pi$, where $d_{\max}$ is a fixed distance scale computed from the environment position bounds. These fixed normalizers keep each potential function stationary during training.

Baseline: Original environment reward with no reward shaping.

Distance potential: encourages movement toward the landing pad, centered at the origin:

$$\tilde{\Phi}_{\text{dist}}(s) = -\min\left(1, \frac{\sqrt{x^2 + y^2}}{d_{\max}}\right). \tag{4}$$

Angle potential – encourages upright lander orientation:

$$\tilde{\Phi}_{\text{angle}}(s) = -\frac{|\theta|}{\pi}. \tag{5}$$

Combined potential – weighted sum of distance and orientation terms:

$$\Phi_{\text{comb}}(s) = 0.7\,\tilde{\Phi}_{\text{dist}}(s) + 0.3\,\tilde{\Phi}_{\text{angle}}(s). \tag{6}$$

All algorithms use SB3 default hyperparameters; any tuning is reported explicitly. Experiments run for 1,000,000 timesteps across 3–5 random seeds. Performance is assessed by mean evaluation reward, success rate (return $\geq$ 200), and sample efficiency (timesteps to first reach 200).

References

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